When Can a Swarm Be Audited?[1]

Evidence limits and an evaluation protocol

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With
Apart Research

Abstract

Multi-agent oversight depends on records that establish which agent acted, what it did, and how work passed between agents. A plausible account can remain unsupported when those relations are missing. This scoping study asks which research direction offers a defensible, feasible contribution to swarm alignment within a three-day sprint. We examine the current sprint materials, selected winning submissions, incident investigations, and primary work on trace integrity, forensic reconstruction, identity and AI control. The synthesis favors a focused oversight experiment on coordinated selective evidence loss and identity churn, rather than another general-purpose logger. Closely related studies already establish trace-tampering risks, evidence-binding failures and useful abstention baselines, narrowing the proposed novelty substantially. An elementary indistinguishability argument shows why stronger reasoning alone cannot recover an actor identity erased from every available record. We specify a prospective experiment that separates instrumentation gains from monitor reasoning using identical evidence packets, clean controls, a deterministic comparator and episode-level uncertainty. Its primary outcome is unsupported attribution, paired with answerability and coverage. No agent-performance experiments are reported. The contribution is a source-grounded problem definition, an explicit evidential limit and a falsifiable protocol for testing whether a small set of externally retained records improves multi-agent auditability.

[1] Research conducted at the Swarm Alignment Hackathon, November 2026

1. Introduction

When several agents share work, an investigator may need to establish a violation, identify a responsible instance, and determine whether agents coordinated. These are different claims. A common bad outcome does not by itself establish collusion; an agent-selected name does not establish identity; and an agent's completion message does not establish execution. Group-level risks motivate evaluating these distinctions explicitly [1,2].

We ask: under equal observation budgets, does selective loss of coordination-critical evidence impair supported attribution more than random loss, especially when agent identities change? This is primarily an oversight question, with implications for identity and audit infrastructure. It favors a small causal comparison over a broad product demonstration.

Our contributions are a primary-source scoping synthesis, a bounded observation showing an information limit, and a prospective evaluation design. The synthesis selects a direction; it does not establish experimental effectiveness or research priority. The proposed empirical contribution remains to be tested.

2. Related Work

Trace-integrity research already demonstrates that local agent harnesses can permit trace deletion, including through peer influence, and recommends recording beyond agent control [3]. Consequently, external logging and multi-agent tampering are not new claims here. Work on AI-system identities and agent infrastructure supplies the accountability context [4,5].

Yun et al. distinguish complete-record agreement from findings justified by the visible evidence. Their forensic reconstruction study includes missing record bindings, justified abstention and a deterministic same-packet comparator [6]. Our proposed delta is coordinated selection of missing evidence interacting with identity churn across workers, not the discovery that correct guesses can lack support.

HELIOS studies joint collusion inference from internal activations, an observation surface distinct from this proposed black-box audit [7]. AI Control provides a trusted/untrusted evaluation framing [8]; CaMeLs motivates system-level enforcement [9]. We do not claim to outperform activation monitoring or to provide a secure deployment runtime. Recent BRIDGE work is an additional overlap lead on fault attribution and calibrated intervention [10].

Incident investigations motivate the question without providing a complete scoring oracle. Swarm Traces explicitly describes uncertain execution outcomes, missing chronology and agent-controlled naming [11]. Such records can support qualitative field-coverage analysis; they cannot automatically label successful attacks or responsible actors.

3. Methods

Scoping method. On 6 November 2026 we reviewed the sprint's overview, resources, guidelines, judging information and schedule; followed its suggested readings; compared the attached

template with the current linked template; and inspected three selected winners and their public critiques [12–15]. We prioritized primary sources with direct methodological relevance. A source audit records reading depth, access failures and unresolved overlap. This was targeted scoping, not a systematic review or a representative study of winners.

Selection criteria. We compared four feasible directions: oversight under evidence loss, external enforcement of delegated permissions, compromised-agent spread across network shapes, and cheap-model proxy validity. We considered likely scientific value, closest prior art, availability of checkable labels, and completion risk. The ranking is an author judgment. No empirical evidence supports a probability of winning.

Prospective environment. One coordinator and three workers operate on fake access requests through a local interpreter. The host retains state, immutable instance identifiers and parent relationships. Workers may change display names. A scripted corruption arm supplies known-positive violations; a later LLM attacker would be a separate exploratory condition. Model output never executes as a shell command or touches real services.

Observation design. Cross identity churn with random versus predeclared selective record loss, holding the number of retained records equal. Separately ablate host identity bindings and execution receipts. Compare a plain monitor, an evidence-constrained monitor and a deterministic solver on precisely the same packet. A full-host oracle is an upper bound, not an equal-information baseline. A fixed targeted selector is a bounded stress test, not an optimal adaptive adversary.

Answerability contract. Enumerate compatible histories in the bounded simulator. Attribution is supported only when every history compatible with the visible packet agrees on the relevant actor; otherwise the correct status is insufficient evidence. The host's hidden full record determines what occurred, while this separate contract determines what can be established. Independent hand-checked examples must validate both labels before confirmatory collection.

Analysis plan. The provisional design uses 120 independent source episodes, balanced between clean and scripted-violation cases, after 20 excluded pilots. Multiple packets and repeated monitor calls remain clustered within episode. Reserve a scenario family for holdout. The primary contrast is targeted minus random unsupported-attribution rate, and its interaction with churn. Use paired episode-bootstrap intervals; report clean false accusations, answerable-case recall, appropriate abstention, definite-answer coverage, invalid outputs and cost. Final models, prompts and stopping rules must be frozen before confirmatory inference.

4. Results

Scoping finding. Oversight under selective evidence loss is the preferred direction because its central comparison has checkable ground truth, equal-information baselines and a concrete operational implication. Delegation enforcement is the fallback if reliable answerability labels cannot be implemented promptly. Table 1 separates the proposed contribution from work that already exists.

Table 1. Existing results and the narrower proposed test. These are research-design comparisons, not measured performance.

Prior work	Already established	Proposed extension
Trace tampering [3]	Integrity and completeness matter	Targeted loss × identity churn
Evidence binding [6]	Supported findings differ from correct guesses	Coordinated multi-worker missingness
HELIOS [7]	Joint inference from activation sensors	Black-box preservation tradeoff

Analytical observation. Consider two equally likely histories, H_A and H_B, in which different worker instances perform the same prohibited update. Suppose all records linking the update to either instance are erased, leaving exactly the same observation O. Any forced-choice monitor receiving only O has the same output distribution in both histories. If it chooses A with probability p and B with probability 1—p, its mean correctness is [p + (1—p)]/2 = 1/2. Additional answers cannot improve forced-choice correctness. This is an elementary information bound under the stated equal-prior construction, not a new general theorem about swarms.

Hidden history	Visible packet after erasure	Correct actor
H_A: worker A updates record	Update complete; actor link absent	A
H_B: worker B updates record	Update complete; actor link absent	B

Figure 1. Two histories collapse to identical evidence. Any monitor restricted to this packet must guess or abstain. Retaining a trustworthy actor–operation receipt makes the observations differ. This construction is analytical, not an observed agent run.

A monitor can avoid unsupported blame by abstaining in that construction, but abstaining everywhere is not useful oversight. Coverage and correct definite findings must therefore accompany unsupported-attribution rates. A trustworthy receipt that differs between histories can break this ambiguity; merely repeating an identical agent-written explanation cannot. Whether feasible receipts deliver a practical benefit is an empirical question.

There are no LLM runs, monitor accuracy estimates, fine-tuning results or measured defense gains in this report. The comparative result is the scoping decision and the analytical result is the bounded ambiguity construction. The empirical protocol is prospective.

5. Discussion and Limitations

The intended theory of change is to help deployers preserve the evidence needed for defensible oversight before an incident. Equal-packet comparisons determine whether a gain comes from reasoning; instrumentation ablations determine whether it comes from better evidence. A useful negative result would identify cases where additional monitor sophistication cannot substitute for missing trusted records.

Novelty is narrow and uncertain. Existing work already covers independent tracing, completeness, evidence binding, abstention and dependency-aware diagnosis. The proposed contribution must demonstrate a distinctive multi-agent interaction or a useful preservation tradeoff beyond those baselines. A null result should narrow the claim rather than be hidden by extra features.

The design assumes a trusted host, bounded histories and a correctly implemented observation contract. Host compromise, off-gateway communication or omitted tool channels invalidate the receipt boundary. Instance provenance does not establish intent or causal mental influence. Simulated tasks and instructed attacks do not establish the prevalence of spontaneous collusion. The scoping review is selective and several linked papers were access-restricted.

Next steps are to validate the toy oracle, freeze the primary protocol and run a small cost pilot. Only then should confirmatory episodes or optional monitor training begin. Sponsor credits are a resource constraint, not evidence of model quality or guaranteed endpoint access.

6. Conclusion

A defensible weekend project should test the evidence boundary of multi-agent oversight. The best-supported candidate is a controlled study of selective record loss and identity churn with matched observations, clean controls and explicit abstention. This report establishes its rationale and a bounded information limit; empirical effectiveness remains unmeasured.

Code and Data

Research artifacts and prospective protocol: https://github.com/razzyrazzy11/swarm-alignment . No experimental dataset has been collected. The October trusted-agent-sting project is preexisting work; its code and results are not new sprint contributions and have not been imported. Its prospective analysis and reproducibility practices informed this proposal [16].

References

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Appendix A. Limitations and Dual-Use / Ethical Considerations

Scope and validity. This is a scoping study and prospective protocol, with one elementary analytical construction. No empirical safety guarantee follows. Literature coverage and ranking are selective; several organizer-linked papers could not be fully accessed. The assumed trusted host and closed tool interpreter require implementation and verification. A hash chain without an external anchor does not prevent full-history rewriting; even anchored integrity does not establish recording completeness.

Statistical limits. The provisional 60 independent clean episodes cannot substantiate a false-positive rate below 1%. Even zero observed false accusations would give a one-sided 95% binomial upper bound of approximately 4.9%. Repeated monitor calls and derived packets are not additional independent episodes. Model families, task families and attack policies would remain a small convenience sample.

Potential misuse. The study could reveal which records an adversary should suppress or which identity changes confuse oversight. It could also help an operator falsely accuse a peer. Keep attack examples confined to fake-record simulations, publish defensive schemas and aggregate results, and review operational details under Apart's responsible disclosure policy before release. No live services, real personal records or incident exploit payloads are needed.

Evidence ethics. Distinguish attempted actions from successful execution and missing evidence from evidence of innocence. Do not assign intent from names, prose or agreement alone. Record uncertain labels, preserve benign provenance and allow insufficient-evidence outputs. Real incident samples require separate permission and redistribution checks. No human-subject study or operational testing is reported.

Appendix B. Prospective resource and release plan

Use Adaption only after verifying the account's available models, endpoint eligibility and actual quoted cost. A proposed maximum allocation is $25 pilot, $250 core inference, $100 replication and $125 reserve, totaling $500. This is a planning ceiling, not a price quote. No credits have been spent. Optional SFT requires a genuinely separate test set and source-episode splits before augmentation; it is not a prerequisite for the core comparison.

LLM Usage Statement

Codex assisted with source discovery, comparison, experimental planning and drafting this report. Primary-source links and access limitations are recorded in the accompanying audit. The author has not yet independently reviewed every claim or run the proposed experiment; no such validation is implied. Final scientific claims and submission remain the author's responsibility.